DO NOW — RECALL

Data structures: OCR exam practice

Name: _____ Class: _____ Date: _____

1. When an item is popped from a stack, is the oldest or the most recently pushed item removed first?

2. Queue: which item is removed first?

3. What do left/right columns contain in an array-based BST?

4. What four actions does OCR 1.4.2(c) require?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

The most recently added item (LIFO).

2. Model answer

The earliest added item (FIFO).

3. Model answer

Pointers/indexes.

4. Model answer

Create, traverse, add data, remove data.

By the end, you can…

- apply data-structure knowledge to OCR past-paper contexts
- use precise language for differences between structures
- interpret pointer-based linked-list and BST representations

Vocabulary — know these terms

Recall the term — LIFO

Recall the term — FIFO

Recall the term — pointer

Recall the term — record

Recall the term — tuple

Recall the term — BST

Recall the term — collision

Recall the term — traverse

NEW LEARNING

Read, annotate and explain

Read the structure before answering

What is stored?
What controls order?
Which pointer/index moves next?
Which OCR term earns the mark?

Today the focus is converting your understanding into short, precise exam answers.

Stack wording

OCR questions often test both the operation and the consequence.

Queue wording

State the ordering difference clearly rather than saying only 'they work differently'.

Linked-list wording

OCR June 2023 explicitly tests how the second item is accessed using pointer values.

BST table wording

OCR examiners reported that candidates often incorrectly put data rather than pointer indexes in left/right columns.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Turn understanding into a 2-mark comparison

Name stack rule

```
Stack removes the most recently added item first.
```

Name queue rule

```
Queue removes the earliest added item first.
```

Link the two

```
Therefore the same input order produces different removal orders.
```

Use exam terms

```
LIFO versus FIFO
```


WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Improve the answer
CODE:
An array has a fixed size once created, whereas a list can grow or shrink.

Guided 2 — CHECK

Improve the answer
CODE:
A tuple differs because ...
CODE:
A tuple is fixed / cannot be changed in the same way, whereas a list is mutable and can change.

Guided 3 — CHECK

Improve the answer
CODE:
Use the ? pointer to locate the first item, then ...
CODE:
Use the start pointer to locate the first item, read its next pointer, then use that pointer value as the index of the second item.

Guided 4 — CHECK

Improve the answer
CODE:
Left/right contain ...
CODE:
Left/right contain pointer/index values to child rows; leaf pointers are null.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

In OCR's June 2024 BST table, the left and right columns contain pointer/index values rather than copies of town names.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY: Array vs list

Give one clear difference.

```
An array and a list store multiple values, but ...
```

Method / working:

Guided 2 — TRY: Tuple vs list

State one difference.

```
A tuple differs because ...
```

Method / working:

Guided 3 — TRY: Linked-list pointer

Explain how to reach the second item.

```
Use the ? pointer to locate the first item, then ...
```

Method / working:

Guided 4 — TRY: BST pointer table

What belongs in left/right columns?

```
Left/right contain ...
```

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Compare structures
Answer the four short questions comparing stack/queue, array/list, tuple/list, and linked-list pointer access.

Question 2

BST array representation
Complete the Left/Data/Right representation for the Irish towns BST.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Record + BST
Explain why a record suits a task containing an ID, real value and order number; then state two BST characteristics.

Question 4

Create / traverse / add / remove
For linked list, stack, queue, graph, BST and hash table, state how one add or remove operation works.